

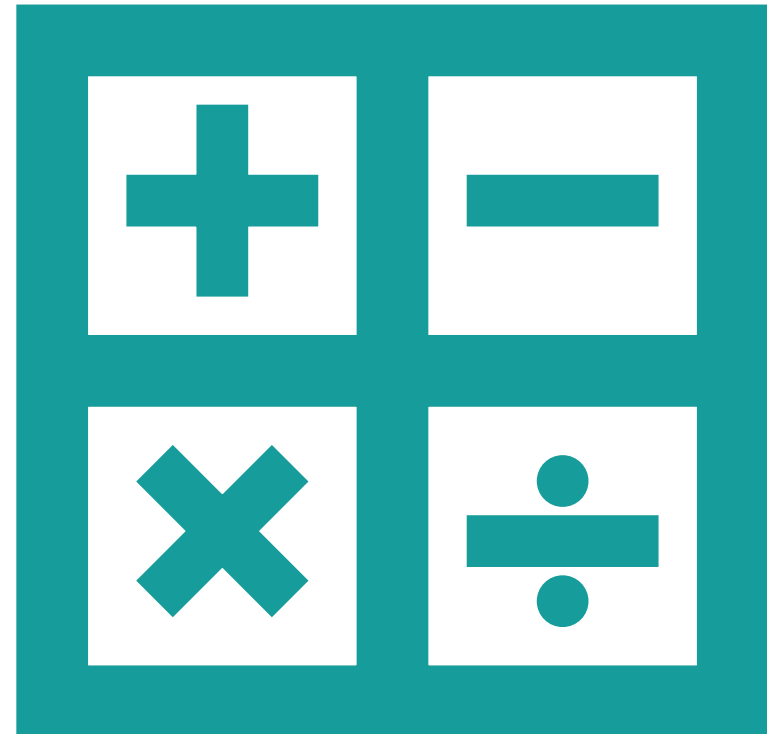


FPGA Simple Calculator

By: Sheraz Javed

Project Overview

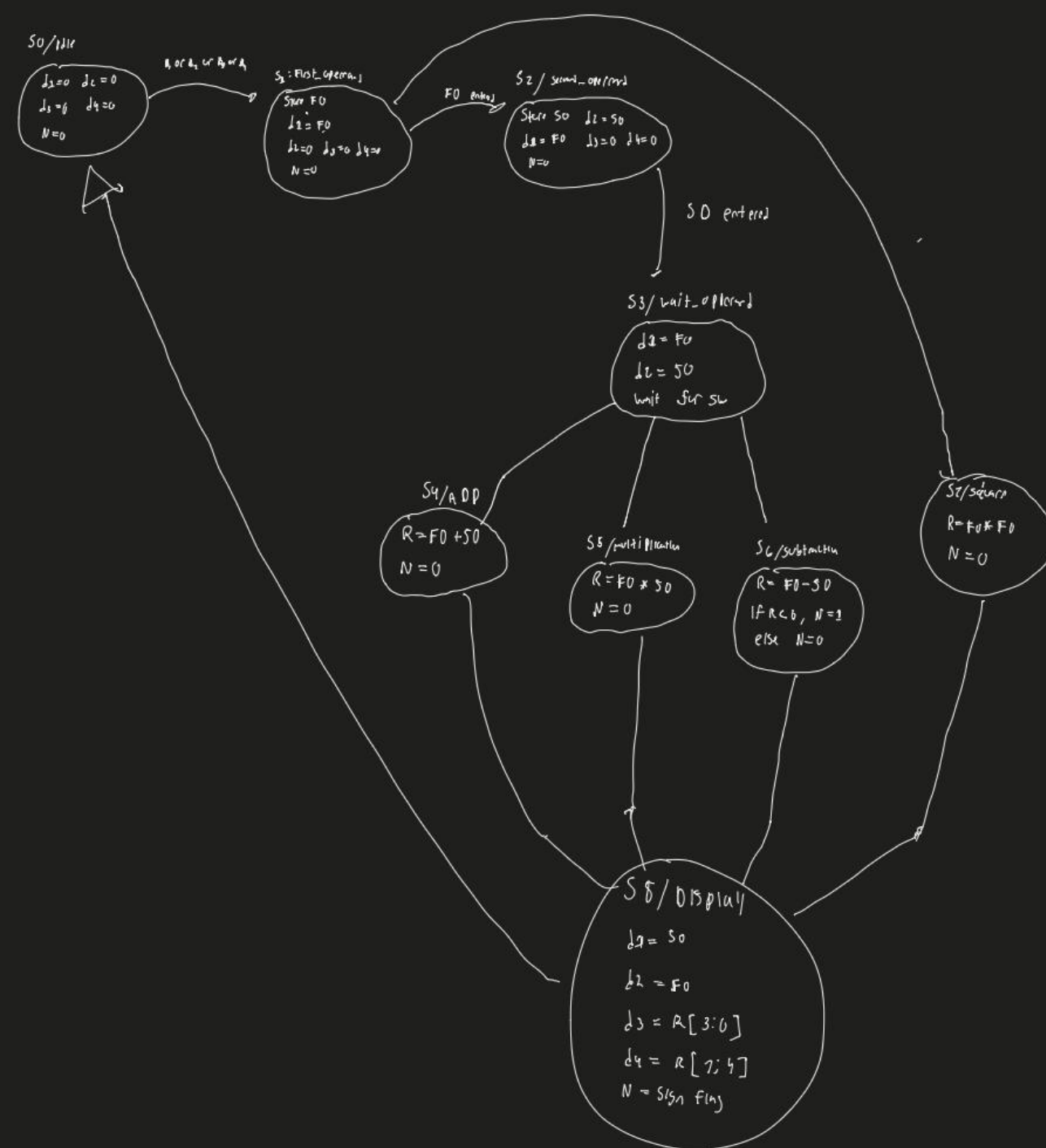
- Operations: addition, subtraction, multiplication, division and Square Function.

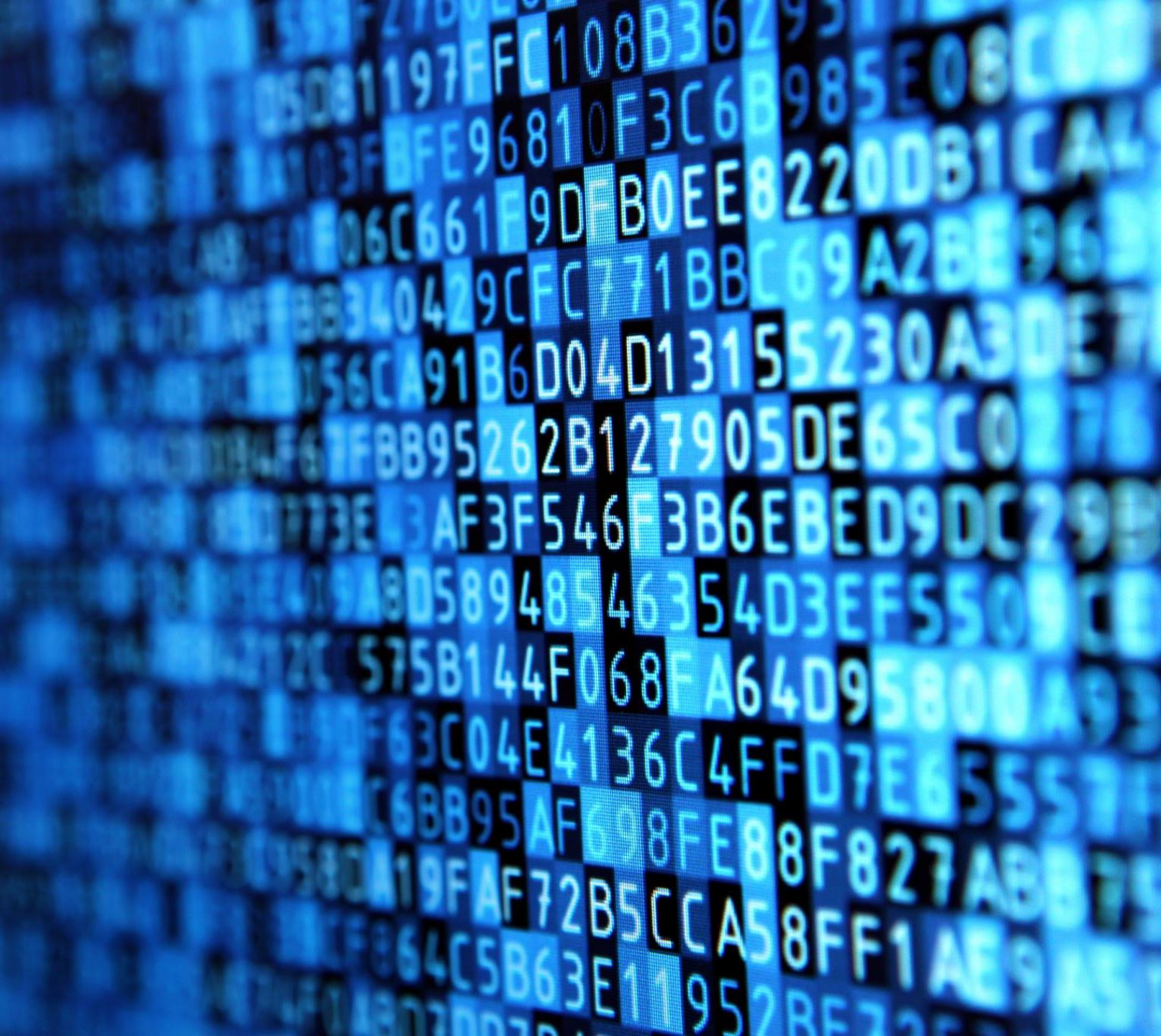


FSM State Diagram

- S0: Idle (reset system, outputs = 0)
- S1: Enter First Operand (FO → disp_1)
- S2: Enter Second Operand (SO → disp_2)
- S3: Wait for operation selection
- S4: Addition → $R = FO + SO$
- S5: Multiplication → $R = FO \times SO$
- S6: Subtraction → $R = FO - SO$ (check sign)
- S7: Division → $R = FO \div SO$
- S8: Display Result (disp_3, disp_4, N flag)

Flow: Idle → FO → SO → Operation → Result → Idle





Input Explanation

- Push buttons mapped to digits 1–5
- Each button converted to 4-bit binary
- Value stored as FO or SO
- Only one button pressed at a time

Verilog Implementation



Microsoft Word
Document

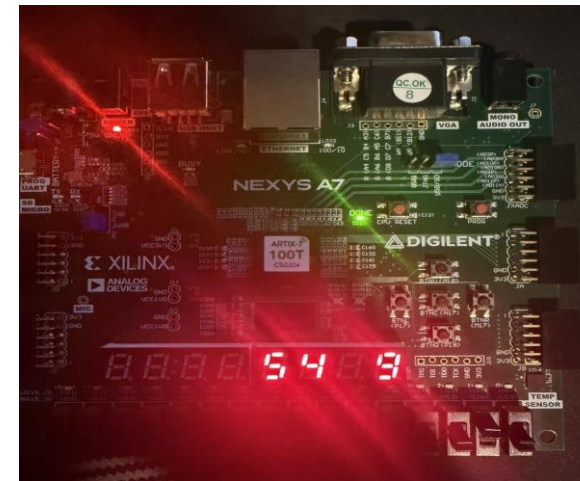
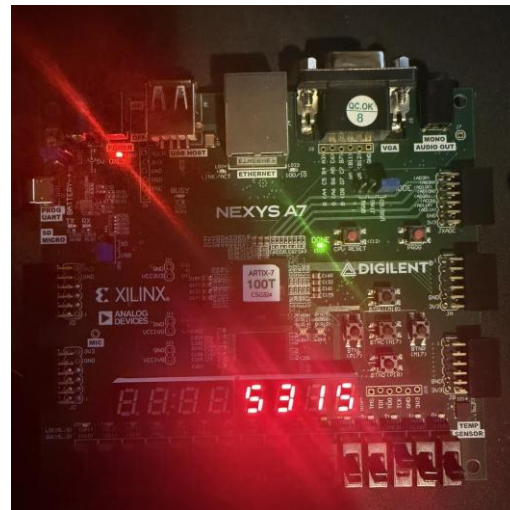
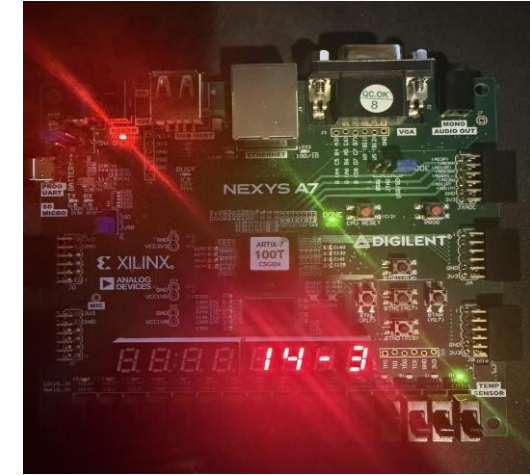
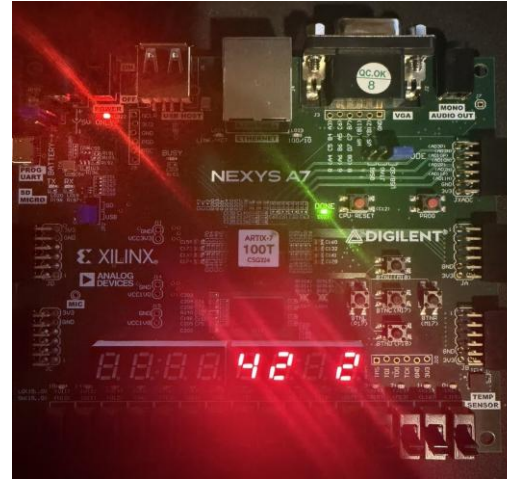
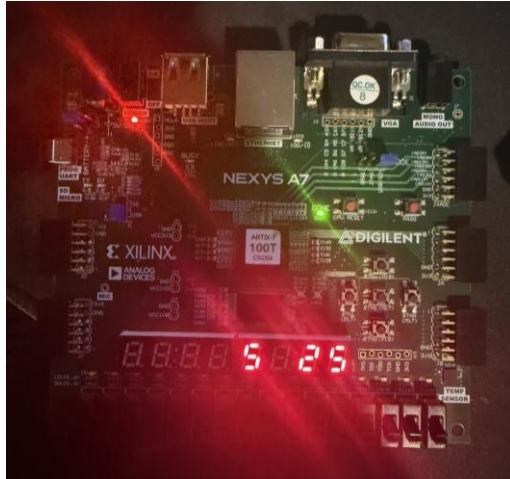
Testbench and Simulation



Name		Value
clk		0
> btn[4:0]		04
> sw[5:0]		00
> seg[6:0]		40
dp		1
> an[7:0]		ff
led_n		0
> PRESS_TIME[31:0]		000001f4
> RELEASE_TIME[31:0]		0000012c
> SW_WAIT[31:0]		000001f4

0 ns	500 ns
00	04
01	00
40	
ff	
000001f4	
0000012c	
000001f4	

Hardware Implementation



Constraints



Microsoft Word
Document

Conclusion

- Implemented an FSM-based calculator on FPGA
- Verified correct operation through simulation and hardware
- Successfully displayed results using 7-segment display
- Learned practical FPGA and Verilog design

